SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – ARCHITECTURE / MODELING (SECURE INFRASTRUCTURE)

Course Code:	CSA5803	**Course Title:**	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2
Weightage:	30% of CIA	**Total Marks:**	100
CO4 Statement:	Design secure DevSecOps infrastructure by developing architecture diagrams, modelling deployment workflows, Identity and Access Management (IAM) relationships, and GitOps workflows.		
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Section / Batch:	Slot c	**Date:**	22/08/26

Instructions to Students

- Design all architecture diagrams, deployment workflows, IAM relationships, and GitOps workflows originally and clearly.
- Use DiagrammeR or Draw.io for architecture modelling and maintain consistent labels, trust boundaries, data flows, and security controls.
- Clearly represent IAM roles/permissions, Infrastructure as Code (IaC), GitOps flow, and DevSecOps security integration wherever applicable.
- Include editable source files for diagrams, IaC/configuration files, methodology documentation, and all supporting project artifacts.
- Submit the GitHub repository link and final PDF report through Google Classroom before the specified deadline; verify that the repository is accessible and complete.

Introduction & IAM

1. Introduction

- DevSecOps integrates development, operations, and security.
- IAM controls user identities and resource access.
- IaC automates infrastructure provisioning.
- GitOps manages infrastructure and application deployment through Git.
- Together, IAM, IaC, and GitOps provide secure and automated infrastructure management.

2. IAM Model

- User Authentication
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Least Privilege Access
- Permission Management
- Audit Logging

IAM Roles:

- Administrator
- Developer
- DevOps Engineer
- Security Engineer
- Auditor

Infrastructure as Code (IaC)

3. IaC Workflow

- Infrastructure requirements are identified.
- Terraform/CloudFormation configuration is created.
- IaC files are stored in Git.
- Configuration is validated automatically.
- Security and compliance checks are performed.
- Approved configuration is used for infrastructure provisioning.

IaC Benefits

- Automation
- Consistency
- Reusability
- Version Control
- Faster Provisioning
- Reduced Configuration Errors

Git and Security Pipeline

4. Git Workflow

Developer
↓
Write Code / IaC
↓
Git Repository
↓
Pull Request
↓
Code Review
↓
Approval

5. Security Validation

- SAST scans source code.
- SCA checks software dependencies.
- IaC scanning identifies insecure configurations.
- Testing verifies application functionality.
- Failed security checks stop the deployment.
- Approved changes move to the deployment stage.

GitOps Workflow

6. GitOps Process

Git Repository
↓
Pull Request
↓
Review & Approval
↓
CI/CD Pipeline
↓
Security Validation
↓
GitOps Controller
↓

Kubernetes / Cloud
↓
Application Deployment

- Git acts as the **single source of truth**.
- GitOps controller continuously monitors the repository.
- Desired state is compared with actual infrastructure.
- Changes are automatically synchronized.
- Configuration drift is detected and corrected.

Security Integration & Automation

7. Security Integration

- IAM controls access to infrastructure.
- RBAC limits user permissions.
- MFA provides stronger authentication.
- Security scanning validates code and infrastructure.
- Approval gates prevent unauthorized deployments.
- Secrets are securely managed.
- Audit logs record important activities.

8. Workflow Automation

Authentication
↓
Authorization
↓
IaC Validation
↓
Security Scanning
↓
Approval
↓
GitOps Synchronization
↓
Deployment

↓
Monitoring
This directly addresses the rubric's requirement for **IAM, IaC, GitOps, security validation, approvals, automation, and auditing**.

Monitoring, Audit & Conclusion

9. Monitoring and Auditing

- Centralized logging collects system activities.
- Deployment activities are recorded.
- IAM activities are monitored.
- Security alerts identify suspicious activities.
- Configuration changes are tracked.
- Audit reports support compliance.

10. Conclusion

IAM provides **secure identity and access control**. IaC provides **automated and consistent infrastructure provisioning**. GitOps provides **controlled, version-based deployment and desired-state management**. Integrating all three creates a secure, automated, traceable, and reliable DevSecOps infrastructure workflow.

Final Overall Workflow

USER
↓
IAM + MFA
↓
RBAC / Least Privilege
↓
DEVELOPER
↓
APPLICATION + IaC
↓
GIT REPOSITORY
↓
PULL REQUEST + REVIEW
↓
CI/CD SECURITY CHECKS
↓
SAST + SCA + IaC SCAN

↓
SECURITY APPROVAL
↓
GITOPS CONTROLLER
↓
CLOUD / KUBERNETES
↓
DEPLOYMENT
↓
MONITORING + LOGGING
↓
AUDIT + COMPLIANCE

This **6-page structure** covers all five rubric categories, including the **30% IaC component and 20% each for IAM, GitOps, and security integration**.